

Akshath Reddy

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PROFESSIONAL SUMMARY:

- **Senior Data Platform Engineer** with 5+ years of experience in distributed systems, real-time **data streaming and AI-enabled data pipelines**, leveraging **AWS, GCP, Spark, Kafka** and cloud-native architectures.
- Architected **TB–PB scale pipelines** using Spark, Kafka and Snowflake with **99.9% SLA reliability**.
- Specialized in **event-driven architectures**, distributed systems and performance optimization (30–50% improvements).
- Hands-on experience building **LLM-powered** data validation and automation frameworks using **AWS Bedrock and LangChain**.
- Proven ability to **design scalable data platforms, optimize cloud costs (up to 50%) and improve system observability**.
- Designed event-driven **data platform architectures** integrating Kafka, Spark, and cloud services for scalable real-time analytics.

TECHNICAL SKILLS:

- **Cloud Platforms:** AWS (S3, Lambda, Kinesis, SQS, Glue, CodePipeline, CloudWatch), Azure (Azure Synapse, Azure Databricks, Event Hubs), Azure Kubernetes Service (AKS), Cloud-Native Architectures
- **Programming Languages:** Java (Multithreading & Concurrency), Python (Data & Backend Development), Scala, SQL, Shell Scripting
- **Data Engineering / Big Data:** Apache Spark (PySpark), Apache Airflow, ETL/ELT Pipeline Design, Snowflake, Azure Synapse, Amazon Redshift, Databricks, Delta Lake, Apache Iceberg, dbt, Data Modeling, Partitioning & Indexing
- **Streaming & Messaging:** Apache Kafka, Kafka Connect, AWS Kinesis, SQS, Azure Event Hubs, Event-Driven Architecture, Real-Time Data Processing
- **Databases:** PostgreSQL, Amazon Redshift, Azure Synapse, Oracle, MySQL, DynamoDB, MongoDB, Relational & NoSQL Systems
- **Containerization / Orchestration:** Docker, Kubernetes (AKS), Microservices Architecture, RESTful Services
- **CI/CD & Version Control:** Git, GitLab CI/CD, Jenkins, AWS CodePipeline, Automated Build & Deployment Pipelines
- **Monitoring & Observability:** CloudWatch, Prometheus, Grafana, Splunk, Ganglia, Metrics, Logging & Alerting
- **AI / Automation:** LLM-Powered / AI-Assisted Pipelines, AI-Driven Data Validation & Analysis, AWS Bedrock, LangChain, Intelligent Workflow Automation
- **Development Practices:** Agile/Scrum, Thread-Safe Component Design, Synchronization & Immutability, Concurrent Collections, Scalable System Design Principles

Certifications:

AWS Certified Solutions Architect – Associate
Microsoft Azure Data Engineer Associate (DP-203)

PROFESSIONAL EXPERIENCE:

American Express, Phoenix, AZ
Data Platform Engineer

July 2024 – Present

Responsibilities:

- Owned and scaled event-driven **microservices platform** on **AWS** supporting analytics and AI workloads across multi-terabyte datasets with 99.9% SLA reliability.
- Engineered **Spark + Airflow pipelines** processing multi-TB/day, reducing latency by 40% and improving SLA adherence from 92% to 99.5% via **checkpointing and idempotent** processing.
- Engineered **fault-tolerant orchestration** framework using **retry strategies, DLQs and checkpointing**, reducing pipeline failures by 60%.

- Developed **thread-safe** shared libraries using immutability patterns and concurrent collections, **eliminating race conditions** under peak load.
- Built resilient REST APIs handling **1K+ requests/sec**, improving system stability using **rate limiting** and **circuit breaker patterns**.
- Engineered real-time ingestion pipelines using **Kinesis, Lambda and SQS** with **auto-scaling and backpressure handling**, achieving sub-minute latency and ~30% higher throughput.
- Optimized **Redshift** and **PostgreSQL** using **distribution styles, sort keys, indexing strategies**, and **query plan analysis** improving performance by **30–50%**.
- Delivered **cloud cost optimization** initiatives through **compute right-sizing and auto-scaling**, reducing infrastructure costs by up to 50%.
- Built **production-grade LLM-powered** data validation framework using **AWS Bedrock + LangChain**, enabling automated anomaly detection and reducing manual validation effort by 40%.
- Enhanced **observability** through **structured logging, distributed tracing, metrics instrumentation**, and **proactive alerting dashboards**.

Capital One, Mclean, VA

May 2023 – June 2024

Software Engineer

Responsibilities:

- Designed and implemented **CDC-based data warehousing** solution enabling scalable analytics across 20+ datasets and improving data availability for downstream systems.
- Built metadata-driven ingestion framework integrating **S3, Snowflake, Spark** and **Lambda**, reducing dataset onboarding time by **60% (3–5 days → <1–2 days)**.
- Developed asynchronous, non-blocking Java microservices improving system throughput by **35%** and reducing latency under peak load.
- Implemented **Snowflake performance tuning** using **clustering keys, materialized views** and **query profiling** improving performance by **30%**.
- Designed and implemented **real-time streaming pipelines** using **Kafka, Spark Structured Streaming** and **Snowflake (Snowpipe)**, processing **10M+ events/day** with exactly-once semantics.
- Built automated **CI/CD pipelines** enabling **zero-downtime deployments** and **environment promotion strategies**.
- Developed **AI-driven ETL generation framework** using **AWS Bedrock + LangChain**, converting Jira requirements into PySpark pipelines and accelerating development by **40%**.
- Implemented **data lineage and governance** framework improving auditability and compliance across pipelines.
- Developed **hybrid LLM + rule-based validation frameworks** ensuring **deterministic, production-ready pipelines**.
- Established **SLA monitoring, failure classification systems**, and **automated rollback mechanisms**.

American Express, Phoenix, AZ

Aug 2022 – April 2023

Data Engineer

Responsibilities:

- Designed and developed ETL pipelines on **Azure (Azure Data Factory, Azure Databricks, ADLS Gen2)** processing **5–8M records/day** with optimized resource utilization.
- Contributed to migration of legacy ETL workflows to **Azure**, improving system scalability and reducing operational overhead by 30%.
- Built event-driven ingestion pipelines using **Azure Event Hubs** with retry/backoff strategies and dead-letter handling.
- Optimized Spark jobs on **Azure Databricks** using partition pruning, broadcast joins, and memory tuning, improving processing efficiency by 25%.
- Implemented data validation frameworks, reconciliation audits, and automated SLA alerts using **Azure Monitor and Log Analytics**.
- Containerized workloads using Docker and deployed on **Azure Kubernetes Service (AKS)** with rolling deployment strategies.

- Reduced **Azure Synapse / ADLS query costs** through partitioning strategies and cost-aware query optimization.

Apollo Health Care Center, Hyderabad, India

July 2019 – Aug 2021

Software Engineer

Responsibilities:

- Developed Kafka-based backend services handling **~3.5M events/day** with reliable offset management and fault-tolerant processing.
- Developed PySpark pipelines processing **~15TB/day**, contributing to ETL SLA reduction from 6 hours to under 1 hour through parallelization and optimization.
- Reduced ETL **SLA from 6 hours → <1 hour** through **parallelization, broadcast joins, and cluster tuning**.
- Contributed to **Oracle → Redshift migration (40TB)**, implementing distribution keys, compression strategies and validation scripts for data accuracy
- Designed **S3-based Data Lake architecture** using **Parquet, compression standards, and lifecycle policies**.
- Implemented **monitoring dashboards, alerting systems**, and conducted **root cause analysis (RCA)** for production incidents.

EDUCATION:

University of Central Missouri

Master of Science in Computer Science.